



Frequentist Persuasion

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Motivation

People interpret communication based on past experience.

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RECOMMENDATION FORM FOR PhD PROGRAM ADMISSIONS

Applicant's Name: _____

Recommender's Name: _____

Institution/Position: _____

Please evaluate the applicant in the following categories by checking the appropriate box in each row.

	Below Average ☐	Average ☐	Good ☐	Top 10% ☐	Top 5% ☐	Exceptional (Top 1%) ☐
Intellectual Ability	☐	☐	☐	☐	☐	☐
Research Skills	☐	☐	☐	☐	☐	☐
Motivation & Initiative	☐	☐	☐	☐	☐	☐
Analytical Skills	☐	☐	☐	☐	☐	☐
Work Ethic	☐	☐	☐	☐	☐	☐
Communication Skills	☐	☐	☐	☐	☐	☐
Overall Potential	☐	☐	☐	☐	☐	☐

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The distribution of available data depends on nature **and** sender's strategy.

Outline

Model	of persuasion where beliefs are estimated from data.
Examples	of over- and under- communication relative to KG .
Characterization	of the sense in which $FP \rightarrow KG$.
Existence	of finite receiver-optimal sample sizes .
Conclusion	and related literature.

Model

The Frequentist Persuasion Game

See [here](#).

Formal Environment

Players	Sender and Receiver
States	Θ (Finite)
Actions	A (Finite)
Messages	M (Finite)
Payoffs	$U^R, U^S : A \times \Theta \rightarrow \mathbb{R}$
Prior	$\mu_0 \in \Delta(\Theta)$

Sample Size $N \in \mathbb{N}$

Learning Rule (estimator) $\ell : (\text{Data}) \times M \rightarrow \Delta(\Theta)$

1. Sender chooses **experiment** $\sigma : \Theta \rightarrow \Delta(M)$
2. N IID (state, message) pairs are drawn from $\Theta \times M$. Call this **data**

$$H_N = \{(\theta_i, m_i)\}_{i=1}^N$$

3. Receiver sees H_N , and a **payoff-relevant** message m .
4. Receiver forms beliefs according to ℓ and acts.
5. Payoffs realized.

Decision Problems

Receiver

$$a(m; H_N) = \arg \max_{a \in A} \sum_{\theta \in \Theta} U^R(a, \theta) \cdot \ell(\theta; m, H_N)$$

Sender (Frequentist)

$$\max_{\sigma} \underbrace{\sum_{\Theta} \sum_M \sum_A \mathbb{P}_{\mu_0}[\theta] \cdot \mathbb{P}_{\sigma}[m|\theta] \cdot \mathbb{P}_{H_N}[a|m] \cdot U^S(a(m; H_N), \theta)}_{V_N(\sigma; \mu_0)}$$

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Sender (Bayesian)

$$\max_{\sigma} \underbrace{\sum_{\Theta} \sum_M \sum_A \mathbb{P}_{\mu_0}[\theta] \cdot \mathbb{P}_{\sigma}[m|\theta] \cdot U^S(a(m), \theta)}_{V^{KG}(\sigma; \mu_0)}$$

Sender's Control: Distribution over Posteriors

$$\Theta = A = M = \{0, 1\}$$

$$U^S = a$$

$$U^R = \mathbb{1}\{a = \theta\}$$

See [here](#).

Examples

Contract Curve for the Binary Case

See [here](#).

Characterizing Convergence of $FP \rightarrow BP$

Definition

A function sequence $V_N : X \rightarrow \mathbb{R}$ *hypo-converges* to $V : X \rightarrow \mathbb{R}$ iff for all $x \in X$:

1. For **every** convergent $x_N \rightarrow x$, $\limsup V_N(x_N) \leq V(x)$.
2. For **some** convergent $x_N \rightarrow x$, $\liminf V_N(x_N) \geq V(x)$.

In this case we write $V_N \xrightarrow{h} V$.

Theorem 1: $FP \xrightarrow{h} BP$

Take arbitrary $(N, \Theta, \mu_0, U^R, U^S)$ and consistent and asymptotically normal ℓ .

Theorem

Let payoffs satisfy Assumption 3 in *Best and Quigley (2023)*.

Then $V_N(\sigma) \xrightarrow{h} V^{KG}(\sigma)$.

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Then $V_N(\sigma) \xrightarrow{h} V^{KG}(\sigma)$.

$$\implies z_N \rightarrow z^{KG} \text{ and } \max V_N \rightarrow \max V^{KG}$$

by Theorem 7.33 in *Rockafellar and Wets (1998)*

Proof Sketch: Intermediate Lemmas

Lemma

$V_N(\sigma) \rightarrow V^{KG}(\sigma)$ whenever σ never leaves a Bayesian receiver indifferent.

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Depends on the Arg-Max CMT of [van der Vaart \(1998\)](#), to show that $\mu_N \rightsquigarrow \mu$ and $a(\mu)$ unique $\implies a(\mu_N) \xrightarrow{P} a(\mu)$.

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Actions are M -Estimators

$$a(\mu_N) = \max_{a \in A} \sum_{\theta \in \Theta} U^S(a, \theta) \cdot \mu_N(\theta)$$

Proof Sketch: Hypo-Convergence

Need two inequalities:

- $\limsup$: u.s.c. + bounded $\implies$ by portmanteau

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Need two inequalities:

- $\limsup$: u.s.c. + bounded $\implies$ by portmanteau
- $\liminf$: $|V_N(\sigma_N) - V^{KG}(\sigma_N)| + |V^{KG}(\sigma_N) - V^{KG}(\sigma)| \rightarrow 0$

Optimal Sample Sizes: The Value of Small Data

$$V_N(z) = \underbrace{\mu_0 + (1-z)\mu_0}_{\pi(z)} \cdot \underbrace{\Pr[\mu_N(z) \geq \frac{1}{2}]}_{c_N(z)}$$

$$V_N(z) = \overbrace{\pi(z)}^{\mu_0 + (1-z)\mu_0} \cdot \overbrace{c_N(z)}^{\Pr[\mu_N(z) \geq \frac{1}{2}]}$$
$$\max_{[z^{KG}, z^{KG} + \varepsilon]} V_N(z) \leq \max_{[z^{KG}, z^{KG} + \varepsilon]} \pi(z) \cdot \max_{[z^{KG}, z^{KG} + \varepsilon]} c_N(z)$$

$\Rightarrow \{z_N\}$ is eventually increasing, and therefore has a minimum.

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Welfare and Extensions

See [here](#).

Conclusion and Related Literature

Related Literature

Bayesian Persuasion	Kamenica and Gentzkow (2011)
Misspecified Persuasion	de Clippel and Zhang (2022) ; Eliaz et al. (2021) ; Dworczak and Pavan (2022)
Noisy Persuasion	Le Treust and Tomala (2019) ; Tsakas and Tsakas (2021)
Persuasion by Experiment	Ball and Espin-Sanchez (2021) ; Patil and Salant (2023) ; Kolotilin et al. (2017)
Persuasion with Learning	Lin and Chen (2024)
Calibrated Forecasting	Jain and Perchet (2026)

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Result 1A Actions in SADPs converge when Bayesian DMs aren't indifferent

Result 2 **More data is not always better for consumers**

Result 2 Frequentist friction **cuts both ways**

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